

# The Economics of Brexit

## Lecture 3: The Single Market

Dr David Chivers

### Macroeconomic Applications and Policy

# Context

One of the main benefits of the Single Market is that firms have access to a **bigger market**.

A larger market means more competition between firms, lower prices and increased consumer welfare.

# Learning objectives

At the end of this topic, you will be able to:

- Analyse how increased market size affects:
  - Competitiveness: the **pro-competitive effect**.
  - The scale of firms: **industrial restructuring**.
  - Welfare.
- Discuss the empirical evidence on the effects of the Single Market Act on European price-cost margins.

# Analytical framework

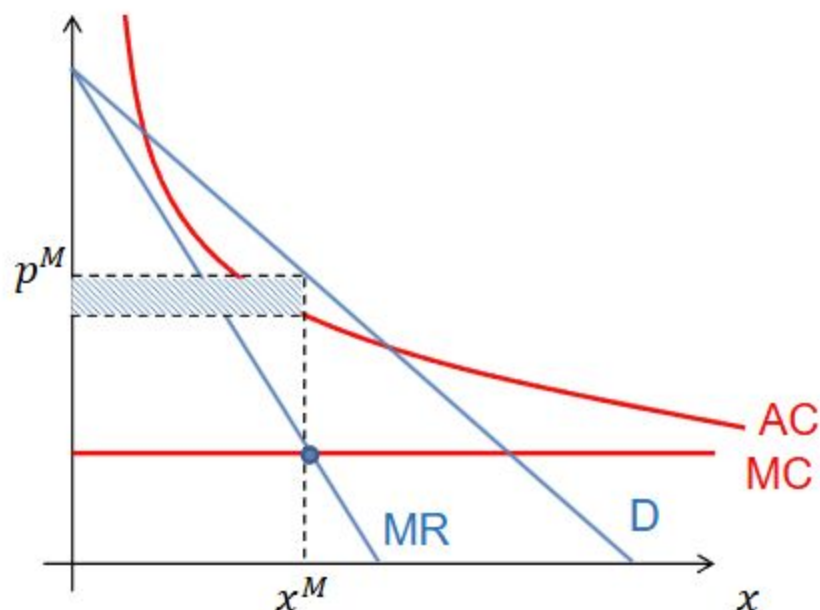
## Brander and Krugman (1983)

- One good.
- Two countries: Home and Foreign.
- A fixed entry cost and constant marginal cost: increasing returns to scale.
- Linear demand.

## Two versions

1. **Fixed number of firms:** one per country.
  - Autarky: two monopolists.
  - Open economy: Cournot competition.
2. **Free entry:** the number of firms is determined endogenously.

# Two monopolistic markets



## Monopolist's problem

$$\max_x px - (cx + F), \quad p = a - bx.$$

## Solution

$$x^M = \frac{a - c}{2b}, \quad p^M = \frac{a + c}{2}.$$

$$\pi^M = \frac{(a - c)^2}{4b} - F.$$

# The two markets integrate

Each firm can sell in both markets. Demand in each country is met by the domestic or foreign firm:

$$Z = x + y \quad (\text{Home demand})$$

$$Z^* = x^* + y^* \quad (\text{Foreign demand})$$

Here,  $x$  and  $x^*$  are Home-firm sales;  $y$  and  $y^*$  are Foreign-firm sales.

## “Iceberg” transport costs

The **marginal cost of exporting** is  $c/g$ , where  $0 < g \leq 1$ .

# Oligopoly: the Home firm's problem

Home firm

$$\max_{x, x^*} px + p^* x^* - cx - \frac{c}{g} x^* - F$$

subject to

$$\begin{aligned} p &= a - bZ = a - b(x + y), \\ p^* &= a - bZ^* = a - b(x^* + y^*). \end{aligned}$$

Substitute the demand functions into the profit function and calculate the **two first-order conditions**.

# Oligopoly: first-order conditions

## Home firm

$$x: a - 2bx - by - c = 0 \quad (1)$$

$$x^*: a - 2bx^* - by^* - \frac{c}{g} = 0 \quad (2)$$

## Foreign firm

$$y: a - 2by - bx - \frac{c}{g} = 0 \quad (3)$$

$$y^*: a - 2by^* - bx^* - c = 0 \quad (4)$$

The Foreign firm's conditions follow from its corresponding maximisation problem.

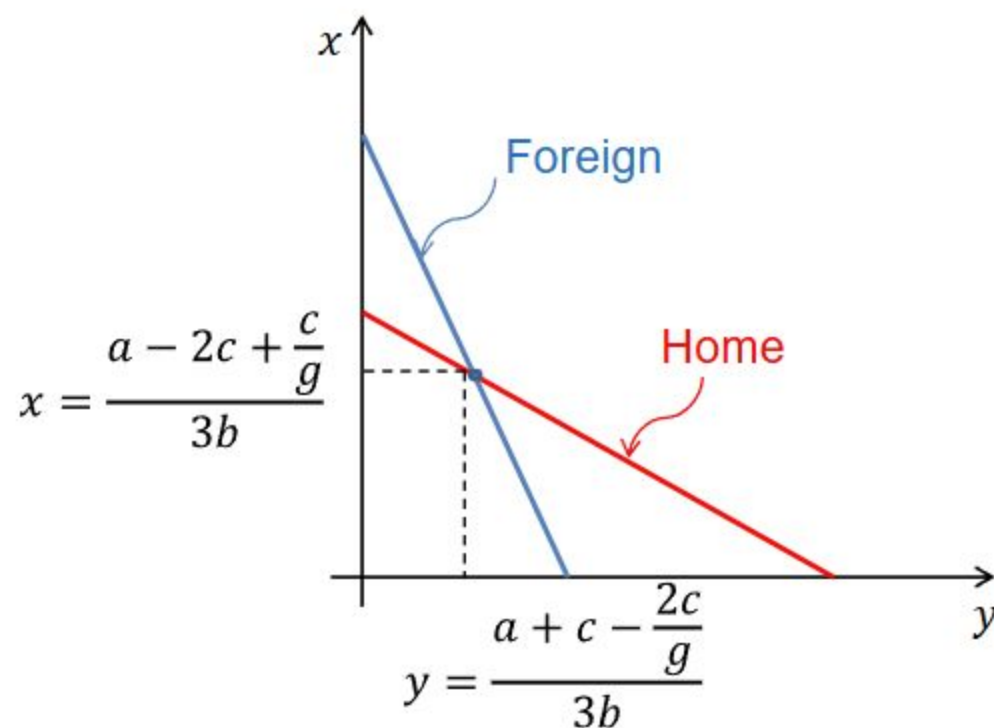
# Oligopoly: reaction functions

The problem is symmetric, so focus on the **Home market**.

Rewrite (1) and (3) to obtain the two firms' reaction functions:

$$x = \frac{a - by - c}{2b}, \quad y = \frac{a - bx - c/g}{2b}.$$

# Equilibrium in the Home market



Notice that  $x > y$  when transport costs are positive.

By symmetry,

$$x^* = y, \quad y^* = x.$$

Each firm takes a larger share of its domestic market because it faces no transport costs there.

# “Reciprocal dumping”

Define Home’s mark-up in each country:

$$\mu \equiv p - c, \quad \mu^* \equiv p^* - \frac{c}{g}.$$

**Proposition:** with positive transport costs, the domestic mark-up is higher than the foreign mark-up.

**Proof:** the countries are identical, so  $Z = Z^*$  and  $p = p^* \equiv p^{FT}$ . Since  $g < 1$ , it follows that  $\mu > \mu^*$ .

# Consequences of international trade

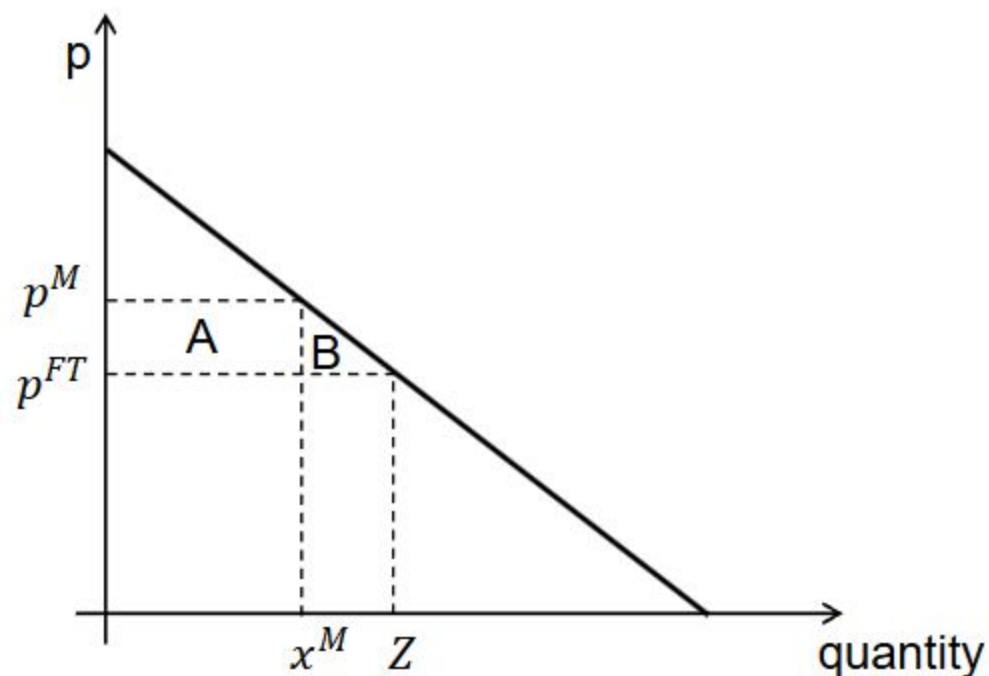
## Question 1

Is there a pro-competitive effect?

## Question 2

If so, does it increase welfare in all countries?

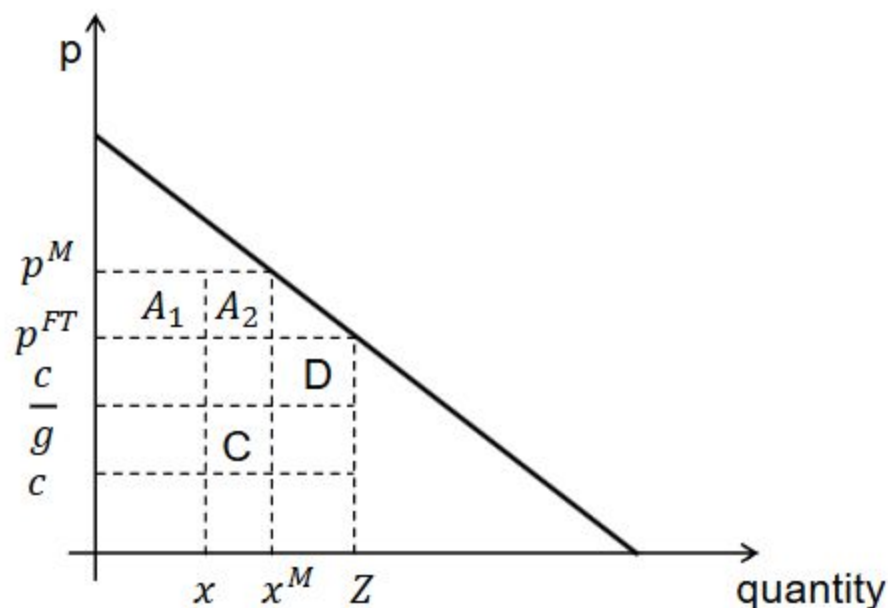
# Welfare effects: consumer surplus



$$\Delta CS = A + B$$

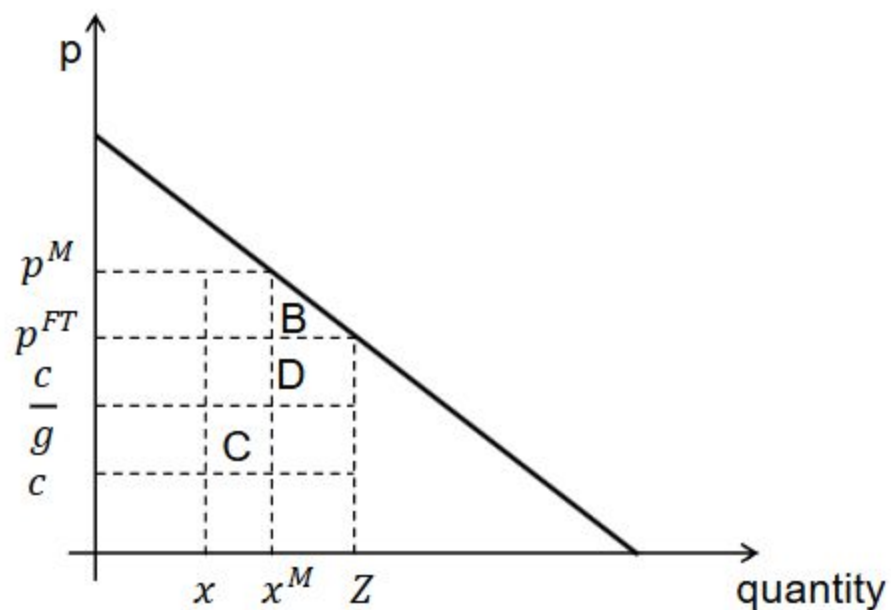
Consumers can consume a larger quantity at a lower price under free trade: the **pro-competitive effect**.

# Welfare effects: producer surplus



- Fall in the mark-up on monopoly sales:  $-A$ .
- Further fall in the mark-up on monopoly sales now made in the foreign market:  $-C$ .
- Additional mark-up on new sales:  $+D$

# Welfare effects: aggregate



**Overall welfare change**

$$B + D - C$$

**What if there are no transport costs?**

# Market size and industrial restructuring

## Questions

- What is the effect of an increased market size on the industry itself?
- Can some firms leave the market because of the increased competition?
- If so, is this beneficial for the countries involved?

# Market size and industrial restructuring

- Brander and Krugman's model (1983) with **free entry** provides an answer.
- The free entry condition allows us to pin down the **number of firms** in the market and therefore to understand the effect of trade on the number of firms.
- First, we look at the closed economy model.

# The COMP curve

Each firm takes the number of firms in the market as given.

The profit-maximisation problem shows that the more firms there are, the less each firm produces (see the appendix).

There is a **negative relationship** between the number of firms  $n$  and the mark-up  $\mu$ .

# The BE curve

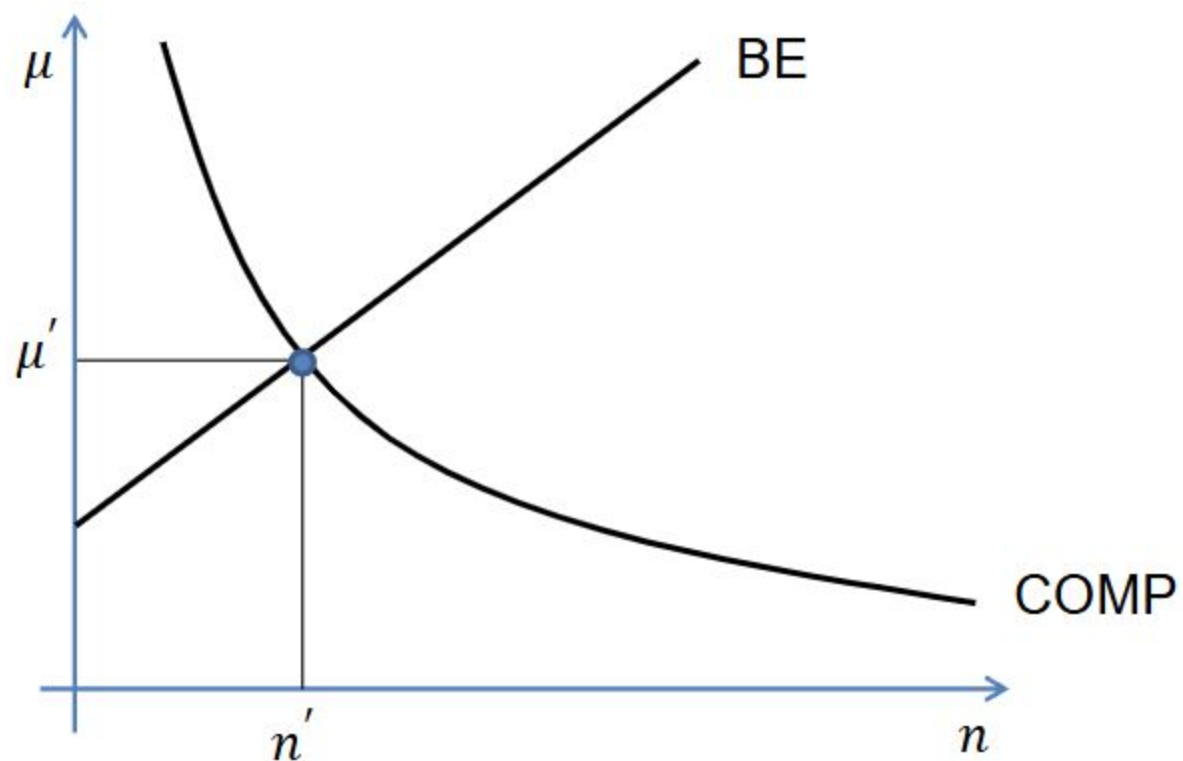
Free entry implies zero profits in the long run:

$$(p - c)x - F = \mu x - F = 0.$$

As  $n$  rises,  $x$  falls. A firm therefore needs a higher  $\mu$  to cover  $F$ .

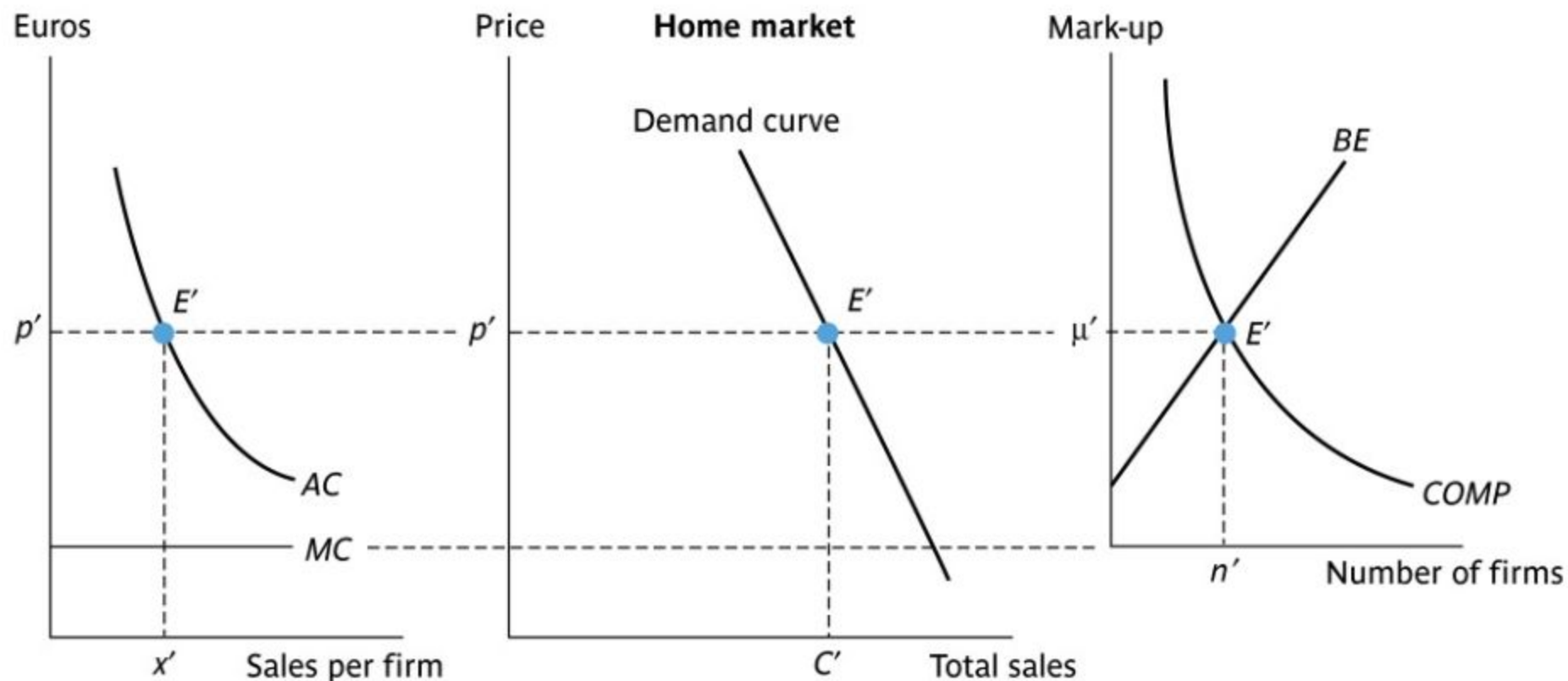
There is a **positive relationship** between the number of firms  $n$  and the break-even mark-up  $\mu$ .

# Equilibrium

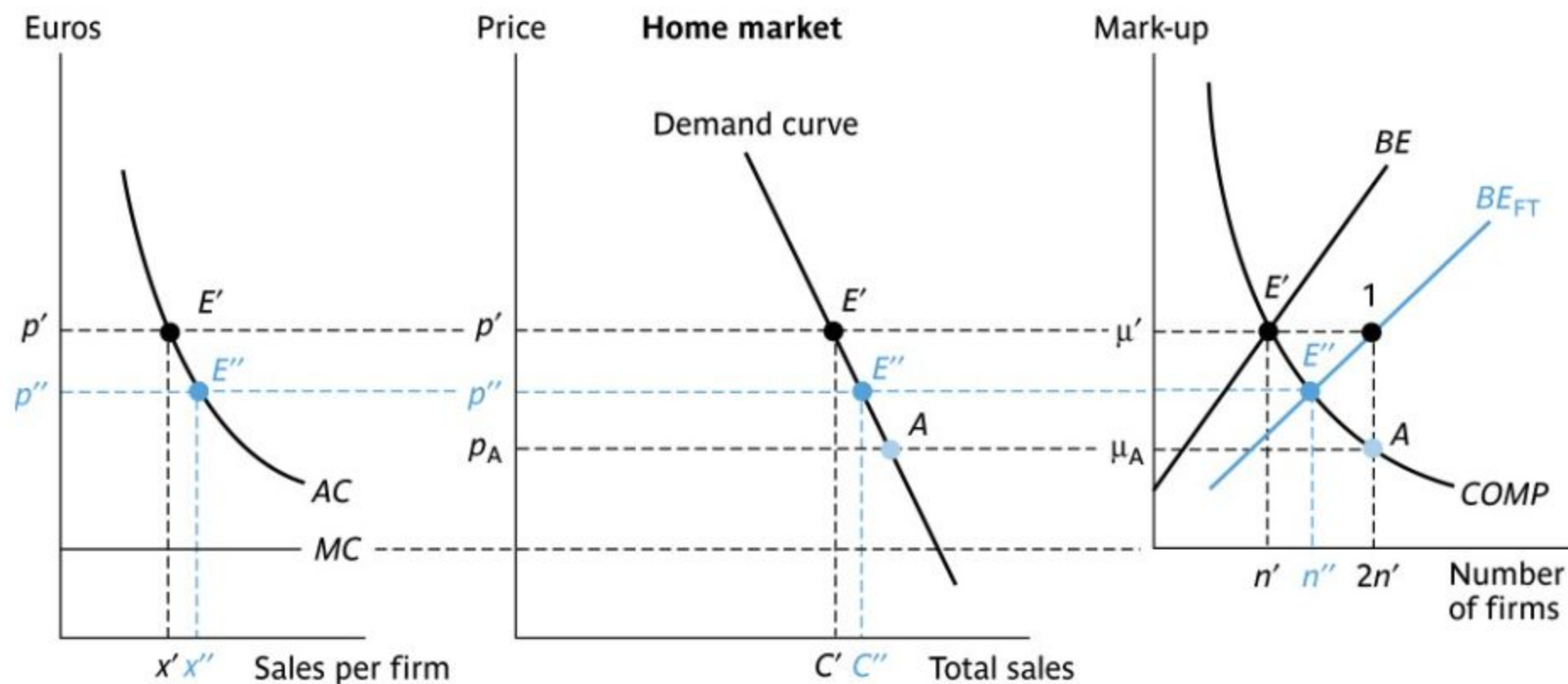


See the appendix for a derivation.

# Before market integration



# After market integration



# EU: a bigger market

## Pro-competitive effect

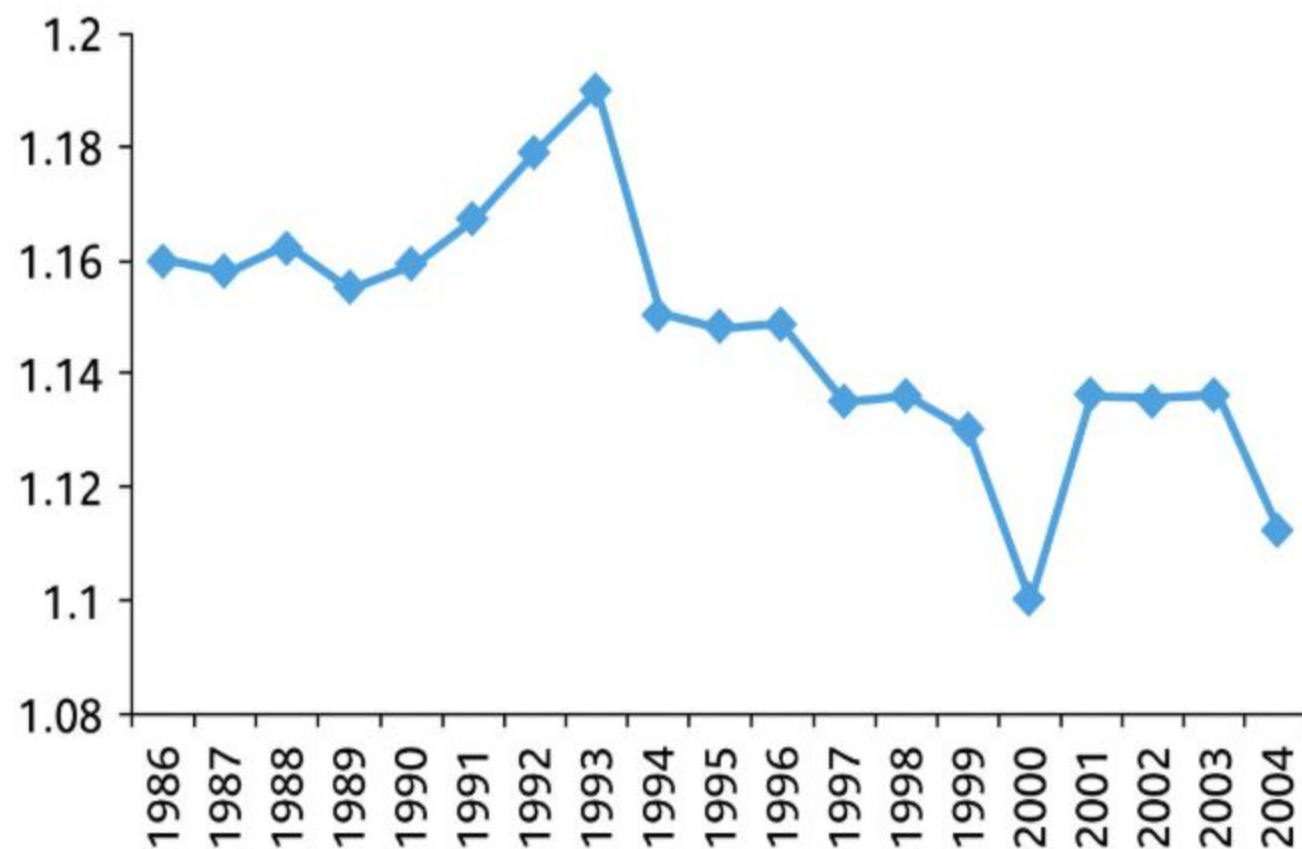
The mark-up falls and output rises because twice as many firms compete in the market.

## Industrial restructuring

- Some firms exit because point A is not feasible:  $P < AC$ .
- The remaining firms are larger and exploit increasing returns to scale more effectively.

**Trade liberalisation is always welfare-improving with free entry in this model. Why?**

# Empirical evidence



Price-cost margins for French industry, **1986–2004**.

Source: Baldwin and Wyplosz (2012).

# Controversies

Trade liberalisation is beneficial for consumers, but not necessarily for firms:

- Profits normally fall.
- Some firms exit the market (job losses).

As a result:

- The incentives to collude can increase as the remaining firms are bigger.
- Governments might be tempted to provide subsidies to national firms.

# Summing up

- Free mobility of goods is beneficial as consumers benefit from the higher competition triggered by the increased market size.
- Market integration also leads to a process of industrial restructuring (job losses if labour mobility across sectors and countries is not perfect).

# Next lecture

- We will talk about migration.

# References

## Essential reading

- Baldwin R. and Wyplosz C. (2022), The Economics of European Integration, Chapter 6
- Brander J., and Krugman P. (1983), A 'reciprocal dumping' model of international trade, Journal of International Economics, 15, 313-321.

## Further reading

See the [ECON2271 Talis reading list](#), also available through Blackboard.

# Appendix: the firm's problem

Firm  $i$  faces the following maximisation problem:

$$\begin{aligned}\max_{x_i} \pi_i &= (p - c)x_i - F, \\ p &= a - bZ, \quad Z = x_i + \sum_{j \neq i}^n x_j.\end{aligned}$$

Substituting the demand function gives

$$\max_{x_i} \left[ a - b \left( x_i + \sum_{j \neq i}^n x_j \right) - c \right] x_i - F.$$

The first-order condition is

$$a - 2bx_i - b \sum_{j \neq i}^n x_j - c = 0.$$

## Appendix: symmetric firms

Assume symmetric firms:  $x_j = x_i = x$ . Then

$$a - 2bx_i - b(n-1)x_i - c = 0,$$

so

$$x_i = x = \frac{a - c}{b(1 + n)}.$$

Using the demand function and  $Z = nx$ , the market price is

$$p = \frac{a + nc}{1 + n}.$$

## Appendix: the COMP equation

The mark-up for a given number of firms gives the **COMP equation**:

$$\mu \equiv p - c = \frac{a - c}{1 + n}.$$

The mark-up decreases with the number of firms. It approaches zero as the number of firms tends to infinity, as under perfect competition:

$$\lim_{n \rightarrow \infty} \mu = 0.$$

## Appendix: the BE equation

The free-entry condition is

$$\mu x = F.$$

The firm's optimal quantity satisfies

$$x = \frac{a - c}{b(1 + n)}.$$

Substitution gives the **BE equation**:

$$\mu = \frac{Fb(1 + n)}{a - c}.$$

# Appendix: equilibrium

We have two equations in two unknowns:  $\mu$  and  $n$ .

Solving the **BE** and **COMP** equations simultaneously gives the equilibrium mark-up and number of firms:

$$\begin{aligned}\mu' &= \sqrt{Fb}, \\ n' &= \frac{a - c}{\sqrt{Fb}} - 1.\end{aligned}$$